

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Models light beams as straight lines and forms vector equations for straight lines using a suitable origin	AO3.3	M1	Modelling beams of light as straight lines taking the origin as point A:
	Forms correct vector equation for a line. Allow one slip	AO1.1b	A1	$\mathbf{r}_A = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 30 \\ 125 \\ 23 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$
	Forms correct vector equation for second line. Allow one slip	AO1.1b	A1	$= \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 30 \\ 125 \\ 20 \end{pmatrix}$
	Forms equation for two components using 'their' model FT 'their' lines	AO3.4	M1	$\mathbf{r}_B = \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 28 \\ 140 \\ 29 \end{pmatrix} - \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix}$
	Solves 'their' equations correctly FT 'their' lines	AO1.1b	A1F	$= \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 20 \\ 130 \\ 28 \end{pmatrix}$
	Checks with third component and concludes that the beams of light intersect This mark is available only if all previous marks have been awarded	AO2.1	R1	$30\lambda = 8 + 20\mu$ $125\lambda = 10 + 130\mu$ $\lambda = \frac{3}{5} \quad \text{and} \quad \mu = \frac{1}{2}$ $3 + \frac{3}{5} \times 20 = 15$ $1 + \frac{1}{2} \times 28 = 15$ $\therefore$ Intersect
(b)	Evaluates scalar product for 'their' direction vectors. (PI)	AO3.1a	M1	$\begin{pmatrix} 30 \\ 125 \\ 20 \end{pmatrix} \cdot \begin{pmatrix} 20 \\ 130 \\ 28 \end{pmatrix} = 17410$
	Sets up equation to find	AO1.1a	M1	

(c)	angle. (PI) FT only if previous M1 awarded			$\cos \theta = \frac{17410}{\sqrt{30^2 + 125^2 + 20^2} \times \sqrt{20^2 + 130^2 + 28^2}}$
	Obtains correct angle.	AO1.1b	A1	$\cos \theta = \frac{17410}{\sqrt{16925} \times \sqrt{18084}}$ $= 0.9951$ $\theta = 5.6^\circ$
	States appropriate refinement.	AO3.5c	E1	Take account of the width of the beams.
Total 10 marks				

Q2.

(a) (i)

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{bmatrix} 8 \\ -4 \\ -6 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ -6 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 0 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Must see $\overrightarrow{OC} - \overrightarrow{OA}$ in correct components.

$n = 5$

B1

1

(ii)

$$\overrightarrow{BC} = \begin{bmatrix} 3 \\ -2 \\ -6 \end{bmatrix}$$

$\overrightarrow{BC}$ or $\overrightarrow{CB}$ correct

B1

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$$5 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -2 \\ -6 \end{bmatrix} = 5\sqrt{2}\sqrt{49}$$

cos ACB

Correct form of formula using consistent vectors; condone use of θ or a wrong angle and a missing multiple of 5

M1

$$5(3 + 2) = 5\sqrt{2}\sqrt{49} \cos ACB$$

Correct scalar product and moduli.

A1

$$\cos ACB = \cos ACB \frac{5}{\sqrt{2} \times 7} = \frac{5\sqrt{2}}{2 \times 7} = \frac{5\sqrt{2}}{14}$$

AG Must see, or rearrangement

$$\cos ACB = \frac{5}{\sqrt{2} \times 7} \text{ or } \frac{25}{35\sqrt{2}}$$

A1

4

(b) vector equation $\mathbf{r} = \begin{bmatrix} 3 \\ 1 \\ -6 \end{bmatrix} + \lambda \begin{bmatrix} 5 \\ -5 \\ 0 \end{bmatrix}$
 $\mathbf{a} + \lambda \mathbf{d}$

M1

OE

A1

2

(c) (i) $\begin{bmatrix} 3 \\ 1 \\ -6 \end{bmatrix} + \lambda \begin{bmatrix} 5 \\ -5 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 1 \\ p \end{bmatrix}$
Equate vector equations for AC and BD. OE

M1

$$3 + 5\lambda = 5 + \mu$$

$$1 - 5\lambda = -2 + \mu$$

Set up equations and solve for μ ; must find a value for μ

M1

$$\mu = \frac{1}{2}$$

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A1

$$-6 = \mu p \Rightarrow p = -12$$

A1

4

(ii) $\overrightarrow{AB} = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix} \quad \overrightarrow{CD} = \begin{bmatrix} -2 \\ 3 \\ -6 \end{bmatrix}$

Clear attempt to find the vectors of the sides.

M1

$$\overrightarrow{AD} = \begin{bmatrix} 3 \\ -2 \\ -6 \end{bmatrix} \quad \overrightarrow{BC} = \begin{bmatrix} 3 \\ -2 \\ -6 \end{bmatrix}$$

All vectors correct

A1

Find the lengths of the sides, or state they all = $\sqrt{49}$
if all correct.

m1

All sides are of same length, 7;
hence rhombus.

Each side = 7 and conclusion.

Or adjacent sides = 7 and opposite sides are parallel.

A1

4

Alternative:

$$\overrightarrow{AC} \cdot \overrightarrow{BD} = 5 - 5$$

Calculate scalar product of $\overrightarrow{AC}$ and $\overrightarrow{BD}$

(M1)

$= 0 \Rightarrow \overrightarrow{AC}$ and $\overrightarrow{BD}$ are perpendicular

$= 0$ from correct $\overrightarrow{AC}$ and $\overrightarrow{BD}$ and conclusion

(A1)

$\mu = \frac{1}{2} \Rightarrow \lambda = \frac{1}{2} \Rightarrow$ intersection is at midpoint of AC and BD

Find value of λ and attempt to use in argument about
point of intersection

(M1)

Diagonals bisect each other at right angles; hence rhombus,
with all sides equal to 7

Fully correct conclusion. Must show diagonals bisect

(A1)

(4)

[15]

Q3.

(a) $\lambda = -1$

$\lambda = -1$ seen or implied

B1

$\lambda = -1$ verified in all three components

Shown

B1

2

(b) $\pm \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$

$\overline{AB}$ or $\overline{BA}$ correct

B1

$$\mathbf{r} = \overline{OA} + \mu \overline{AB} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + \mu \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$$

$\mathbf{a} + \mu \mathbf{d}$

M1

OE; ft on $\overline{AB}$ or $\overline{BA}$

A1ft

3

(c) $\overline{CD} = \overline{OD} - \overline{OC}$

$$= \begin{bmatrix} 3-2\mu \\ -2-3\mu \\ 4+2\mu \end{bmatrix} - \begin{bmatrix} -4 \\ 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 7-2\mu \\ -7-3\mu \\ 5+2\mu \end{bmatrix}$$

$\pm \overline{CD}$ in terms of μ

OE

B1

$$\overline{CD} \cdot \overline{AB} = 0 \quad \text{or} \quad \overline{CD} \cdot \overline{AD} = 0$$

$$= \begin{bmatrix} 3-2\mu \\ -2-3\mu \\ 4+2\mu \end{bmatrix} \cdot \begin{bmatrix} -4 \\ 5 \\ -1 \end{bmatrix} = 0$$

Candidate's $\overline{CD}$ sp with candidate's $\overline{AB}$ or $\overline{AD} = 0$

PI by a solution for μ

M1

$$-14 + 4\mu + 21 + 9\mu + 10 + 4\mu = 0$$

$$17 + 17\mu = 0$$

$$\mu = -1$$

Expand sp to an equation in μ and solve for μ

m1A1

D is at (5, 1, 2)

Accept as a column vector

A1

Alternative using Pythagoras

$$\overrightarrow{CD} = \overrightarrow{OD} - \mu \overrightarrow{OC}$$

$$= \begin{bmatrix} 3-2\mu \\ -2-3\mu \\ 4+2\mu \end{bmatrix} - \begin{bmatrix} -4 \\ 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 7-2\mu \\ -7-3\mu \\ 5+2\mu \end{bmatrix}$$

$\pm \overrightarrow{CD}$ in terms of μ

(B1)

$$AC^2 = AD^2 + CD^2$$

$$(7^2 + 7^2 + 5^2) = \mu^2(2^2 + 3^2 + 2^2) + ((7 - 2\mu)^2 + (7 + 3\mu)^2 + (5 + 2\mu)^2)$$

$$\overrightarrow{AC} = \begin{bmatrix} -7 \\ 7 \\ -5 \end{bmatrix} \quad \overrightarrow{AD} = \begin{bmatrix} -2\mu \\ -3\mu \\ 2\mu \end{bmatrix}$$

Correct Pythagoras expression in terms of μ ;

(M1)

$$123 = 17\mu^2 + 123 + 34\mu + 17\mu^2$$

Multiply out and solve to find a value for μ

(m1)

$$0 = 34\mu^2 + 34\mu$$

$$\mu = -1 \quad (\mu = 0 \text{ is point A})$$

$$\mu = -1$$

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(A1)

D is at (5, 1, 2)

(A1)

(5)

(d) $\overrightarrow{OE} = \overrightarrow{OA} + \overrightarrow{AE} = \overrightarrow{OA} + 3\overrightarrow{AD}$

Accept $AE = 3AD$

M1

$$\overrightarrow{OE} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix}$$

E is at (9, 7, -2)

Accept as a column vector

A1

OR
 $\overrightarrow{OE} = \overrightarrow{OA} + \overrightarrow{AE} = \overrightarrow{OA} + 3\overrightarrow{DA}$
 Accept $AE = 3DA$

M1

$$\overrightarrow{OE} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix} \quad E \text{ is at } (-3, -11, 10)$$

Accept as a column vector

A1

4

Alternative

$$|\overrightarrow{DE}| = 2|\overrightarrow{AD}| \Rightarrow \overrightarrow{OE} = \overrightarrow{OD} + 2\overrightarrow{AD}$$

(M1)

$$\overrightarrow{OE} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} \quad E \text{ is at } (9, 7, -2)$$

(A1)

$$|\overrightarrow{DE}| = 4|\overrightarrow{DA}| \Rightarrow \overrightarrow{OE} = \overrightarrow{OD} + 4\overrightarrow{DA}$$

(M1)

$$\overrightarrow{OE} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} + 4 \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix} \quad E \text{ is at } (-3, -11, 10)$$

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(A1)

(4)

[14]

Q4.

(a) (i)
$$\overrightarrow{AB} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} - \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ -4 \end{bmatrix}$$

$\pm (\overrightarrow{OB} - \overrightarrow{OA})$ implied by two correct components

M1

Allow as $(-2, 2, -4)$

A1

2

(ii) $\begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix} \bullet \overrightarrow{AB} = -2 + 10 + 8 = 16$

M1

ft on $\overrightarrow{AB}$

A1ft

$$\cos \theta = \frac{16}{\sqrt{24}\sqrt{30}}$$

Correct formula for $\cos \theta$ with consistent vectors and correct moduli, in form $\sqrt{a^2 + b^2 + c^2}$

M1

$$\theta = 53^\circ$$

CSO Accept 53.4° , 53.40°

A1

4

(b) $\overrightarrow{AB} \bullet \overrightarrow{BC} = \begin{bmatrix} -2 \\ 2 \\ -4 \end{bmatrix} \bullet \left(\begin{bmatrix} 4+p \\ -2+5p \\ 3-2p \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \right)$

SC B1 90° following $sp = 0$

Set up scalar product. $\mu = p$ at C. Any letter for p.

M1

$$\overrightarrow{BC} = \begin{bmatrix} 2+p \\ -2+5p \\ 4-2p \end{bmatrix}$$

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Clear attempt to find $\overrightarrow{BC}$ in terms of p.

$\overrightarrow{BC}$ or $\overrightarrow{CB}$ correct

B1

$$-4 - 2p - 4 + 10p - 16 + 8p = 0$$

m1

$$16p = 24 \quad p = \frac{3}{2}$$

Expand scalar product and solve for p;
(= 0 possibly implied)

A1

$$\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{BC} = \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix} + \begin{bmatrix} \frac{7}{2} \\ \frac{11}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{15}{2} \\ \frac{7}{2} \\ 4 \end{bmatrix}$$

Correct vector expression to find $\overrightarrow{OD}$ written in components

m1

$$D \text{ is at } \left(\frac{15}{2}, \frac{7}{2}, 4 \right)$$

CAO; condone column vector

6

A1

Alternative for last 2 marks

$$\overrightarrow{OD} = \overrightarrow{OC} + \overrightarrow{BA} = \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix} + \frac{3}{2} \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}$$

(M1)

$$D \text{ is at } \left(\frac{15}{2}, \frac{7}{2}, 4 \right)$$

(A1)

NB $p = \frac{3}{2}$ can come from wrong working where candidate uses $\overrightarrow{OC}$ in place of $\overrightarrow{BC}$.

This is M0 and scores no further marks, (unless they happen to find and go on to use it correctly).

(6)

[12]

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Q5.

(a) Use of scalar product on

$$\begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} \text{ and } \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix}$$

M1

Sc.Prod. = ± 21 ;

B1

Moduli $\sqrt{54}$ and $\sqrt{26}$ correct

Accept 7.348... & 5.099...

B1

AWRT 56°

A1

- (b) $2x + y + 7t = 10$ and $3x + y - 4t = 7$ noted or used

M1

Eliminating (say) y to get x as a fn. of t

M1

$$x = 11t - 3$$

CAO

A1

Subst^g. back for y

M1

$$y = 16 - 29t$$

CAO

A1

$$\frac{x+3}{11} = \frac{y-16}{-29} = z \quad (=t)$$

B1 ft

6

- (c) Attempt at either $\begin{pmatrix} \lambda+20 \\ 9\lambda-1 \\ 4\lambda+7 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} = 10$ or $\begin{pmatrix} \lambda+20 \\ 9\lambda-1 \\ 4\lambda+7 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} = 7$.

M1

Solving either

$$40 + 2\lambda - 1 + 9\lambda + 49 + 28\lambda = 10 \text{ or}$$

$$60 + 3\lambda - 1 + 9\lambda - 28 - 16\lambda = 7$$

M1

$$\lambda_1 = -2$$

$$\lambda_2 = 6$$

A1A1

$$P = (18, -19, -1) \text{ and } Q = (26, 53, 31)$$

$$PQ = \sqrt{8^2 + 72^2 + 32^2} = \sqrt{6272} = 56\sqrt{2}$$

NB P, Q not required:

$$d = |\lambda_1 - \lambda_2| \times |\mathbf{i} + 9\mathbf{j} + 4\mathbf{k}| = 8 \times 7\sqrt{2} = 56\sqrt{2} \quad \mathbf{M1 \ A1}$$

Q6.

(a) $2\lambda = 8 + 2\mu$
 $-2 = 3 + 5\mu$
 $\lambda = 3, \mu = -1$

Use the first two equations to set up and attempt to solve simultaneous equations for λ or μ . Must not assume $q = 4$.

M1

$q - \lambda = 5 + 4\mu$
 $q = 5 + 3 - 4 = 4$

Use 3rd equation to show $q = 4$ **AG**.

A1

p is at (6, -2, 1)

Condone as a column vector

B1

(b) $\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} = 4 - 4 = 0 \Rightarrow \text{perpendicular}$
 or $2 \times 2 + -1 \times 4 = 0$ seen **and** conclusion
 (condone $\theta = 90$)

B1

1

(c) (i) A is at (2, -2, 3)
 $AP^2 = (6 - 2)^2 + (-2 - -2)^2 + (1 - 3)^2$
 Candidate's $|\overline{AP}|^2$

M1

$= 20$
 CAO
 NMS $AP = \sqrt{20}$ M1A0

A1

2

(ii) $(\overline{PB}) = \begin{bmatrix} 8 \\ 3 \\ 5 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} - \begin{bmatrix} 6 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 + 2\mu \\ 5 + 5\mu \\ 4 + 4\mu \end{bmatrix}$
 Clear attempt to find $\overline{BP}$ or $\overline{PB}$ in terms of μ

M1

$$(PB^2)=(2+2\mu)^2+(5+5\mu)^2+(4+4\mu)^2$$

Find distance BP in terms of μ

m1

$$45\mu^2+90\mu+45=20$$

$$(5)(9\mu^2+18\mu+5)=0$$

Attempt to set up three-term quadratic in μ and equate to their AP^2

m1

$$(3\mu+1)(3\mu+5)=0$$

Solve quadratic equation to obtain **two** values of μ

m1

$$\mu = -\frac{1}{3} \text{ and } \mu = -\frac{5}{3}$$

Both values correct.

A1

$$B \text{ is at } \left(\frac{22}{3}, \frac{4}{3}, \frac{11}{3}\right) \text{ and } \left(\frac{14}{3}, -\frac{16}{3}, -\frac{5}{3}\right)$$

Both sets of coordinates required.

Condone column vectors.

SC one value of μ correct and corresponding coordinates of B correct scores A1 A0.

A1

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Alternative 1:

$$(\overrightarrow{AB}) = \begin{bmatrix} 8 \\ 3 \\ 5 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} - \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6+2\mu \\ 5+5\mu \\ 2+4\mu \end{bmatrix}$$

Clear attempt to find $\overrightarrow{AB}$ or $\overrightarrow{BA}$ in terms of μ

(M1)

$$(AB^2)=(6+2\mu)^2+(5+5\mu)^2+(2+4\mu)^2$$

Find distance AB in terms of μ

(m1)

$$45\mu^2+90\mu+65=40$$

$$(5)(9\mu^2+18\mu+5)=0$$

Attempt to set up three-term quadratic in μ and

equate to their $2 \times$ their AP^2

(m1)

As before

Alternative 2:

$$\overrightarrow{PB} = k \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$$

(M1)

$$k^2(2^2 + 5^2 + 4^2) = 20$$

m1 for LHS

(m1)

m1 for equating to 'their 20'

May score M1m0m1

(m1)

$$k = \pm \frac{2}{3}$$

(A1)

$$\overrightarrow{OB} = \overrightarrow{OP} + (\pm)(\text{their value of } k) \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$$

(m1)

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B is at $\left(\frac{22}{3}, \frac{4}{3}, \frac{11}{3}\right)$ and $\left(\frac{14}{3}, \frac{16}{3}, \frac{5}{3}\right)$

(A1)

6

[12]

Q7.

(a) $\begin{bmatrix} 7 \\ 4 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 14 + 4 + 18 = 36 \Rightarrow Q \text{ in } \Pi$

B1

1

(b) $\begin{bmatrix} -7 \\ 5 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = -14 + 5 + 9 = 0$

Shown

B1

Explanation that l is perpendicular to l 's normal $\Rightarrow l$ is parallel to l

B1

1

Alternative:

$$\begin{bmatrix} 20-7\mu \\ 5\mu-8 \\ 3\mu+1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 35 \quad (+0\mu)$$

(B1)

Explanation that, since this $\neq 36$ (and constant), l does not intersect l and must therefore be parallel to it

(B1)

2

- (c) **Mark according to whichever scheme gives the greatest credit**

MARK SCHEME I

For $\mathbf{p} = \mathbf{q} + \lambda \mathbf{n} = \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix}$ for some λ

B1

For $PQ = |\lambda|\sqrt{14}$ or equivalent

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$$\overrightarrow{PR} = \mathbf{r} - \mathbf{p} = \begin{bmatrix} 20-7\mu \\ 5\mu-8 \\ 3\mu+1 \end{bmatrix} - \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix}$$

M1

$$= \begin{bmatrix} -7\mu-2\lambda+13 \\ 5\mu-\lambda-12 \\ 3\mu-3\lambda-5 \end{bmatrix}$$

A1

Setting $\overrightarrow{PR} \cdot \begin{bmatrix} -7 \\ 5 \\ 3 \end{bmatrix} = 0$

M1

Solving linear equation for λ

$$\Rightarrow 83\mu - 166 = 0 \text{ ultimately}$$

M1

$$\mu = 2 \text{ and/or } \overline{PR} = \begin{bmatrix} -(2\lambda+1) \\ -(\lambda+2) \\ 1-3\lambda \end{bmatrix}$$

A1

$$PR^2 = PQ^2$$

$$(1+2\lambda)^2 + (2+\lambda)^2 + (1-3\lambda)^2 = 14\lambda^2$$

M1

$$(\lambda = -3)$$

$$P = (1, 1, -3)$$

A1

9

MARK SCHEME II

$$\text{For } \mathbf{p} = \mathbf{q} + \lambda \mathbf{n} = \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix} \text{ for some } \lambda$$

B1

$$\text{For } PQ^2 = 14\lambda^2$$

B1

$$\overline{PR} = \mathbf{r} - \mathbf{p} = \begin{bmatrix} 20-7\mu \\ 5\mu-8 \\ 3\mu+1 \end{bmatrix} - \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix}$$

M1

$$= \begin{bmatrix} -7\mu-2\lambda+13 \\ 5\mu-\lambda-12 \\ 3\mu-3\lambda-5 \end{bmatrix}$$

A1

$$PR^2 = (13+7\mu+2\lambda)^2 + (12-5\mu+\lambda)^2 + (5-3\mu+3\lambda)^2$$

M1

$$\text{Setting } PR^2 = PQ^2$$

M1

$$\Rightarrow 83\mu^2 - 332\mu + 338 + 2\lambda = 0$$

Correct quadratic in μ

A1

Considering discriminant = 0 or $83(\mu - 2)^2 = -2(\lambda + 3)$

M1

$$\mu = 2, \lambda = -3 \text{ and } P(1, 1, -3)$$

A1

9

MARK SCHEME III

For $\mathbf{p} = \mathbf{q} + \lambda \mathbf{n} = \begin{bmatrix} 7+2\lambda \\ 4+\lambda \\ 6+3\lambda \end{bmatrix}$ for some λ

B1

For $PQ^2 = 14\lambda^2$

B1

For $\mathbf{r} = \begin{bmatrix} 20-7\mu \\ 5\mu-8 \\ 3\mu+1 \end{bmatrix}$

B1

For $QR^2 = (13 - 7\mu)^2 + (5\mu - 12)^2 + (3\mu - 5)^2$
 $= 83\mu^2 - 332\mu + 338 = 83(\mu - 2)^2 + 6$

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M1

For R closest to Q when $\mu = 2$,

M1

$$R = (6, 2, 7)$$

A1

Then $\overline{RP} = \begin{bmatrix} 1+2\lambda \\ 2+\lambda \\ 3\lambda-1 \end{bmatrix}$
ft

B1

Setting $PR^2 = PQ^2 \Rightarrow (1 + 2\lambda)^2 + (2 + \lambda)^2 + (3\lambda - 1)^2 = 14\lambda^2$

M1

$$(\lambda = -3) \quad P = (1, 1, -3)$$

A1

9

[12]

Q8.

(a) (i)
$$\overrightarrow{AB} = \begin{bmatrix} 6 \\ 0 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$$

 $\pm (\overrightarrow{OB} - \overrightarrow{OA})$ implied by 2 correct components

M1

A1

2

(ii)
$$\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = 6 - 2 - 3 = 1$$

Scalar product with correct vectors; allow one component error.

M1

ft on $\overrightarrow{AB}$

A1ft

$$\cos \theta = \frac{sp}{\sqrt{14}\sqrt{14}}$$

Correct form for $\cos \theta$ with one correct modulus

m1

$$\cos \theta = \frac{1}{14} \quad \theta = 85.9^\circ$$

cs0 85.9 or better

A1

4

(b) (i)
$$\overrightarrow{OD} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 7 \\ -4 \\ 10 \end{bmatrix}$$

 Implied by 2 correct components

M1

$$\text{line } l_2 \quad r = \begin{bmatrix} 7 \\ -4 \\ 10 \end{bmatrix} + \mu \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$$

$$r = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ required ft on } \overline{AB}$$

A1ft

2

$$(ii) \quad \overline{BC} = \overline{OC} - \overline{OB} = \begin{bmatrix} 1+3p \\ -4+2p \\ 7-p \end{bmatrix}$$

 $\mu = p$ at C

Find $\overline{BC}$ in terms of p

M1

$$\overline{AD} = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix} \quad |\overline{BC}| = \sqrt{56}$$

$$PI \quad B1 \text{ is for } |\overline{BC}| = \sqrt{56}$$

B1ft

$$(1+3p)^2 + (-4+2p)^2 + (7-p)^2 = 56$$

m1

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$$14p^2 - 24p + 66 = 56$$

$$7p^2 - 12p + 5 = 0$$

ft on $|\overline{BC}|$

Simplification to quadratic equation with all terms on one side

m1

$$(7p-5)(p-1) = 0$$

$$p = \frac{5}{7} \text{ and } p = 1$$

Exact fraction required

A1

$$C \text{ is at } \left(9\frac{1}{7}, -2\frac{4}{7}, 9\frac{2}{7}\right)$$

cs0 Accept as column vector

A1

6

Alternative : Using equal angles

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{bmatrix} 1+3p \\ -4+2p \\ 7-p \end{bmatrix}$$

$$\mu = p \text{ at } C$$

Find $\overrightarrow{BC}$ in terms of p

(M1)

$$\overrightarrow{AD} = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix} \quad |\overrightarrow{BC}| = \sqrt{56}$$

(B1ft)

$$(\cos \theta) = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{\sqrt{14}\sqrt{56}} = \frac{\begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1+3p \\ -4+2p \\ 7-p \end{bmatrix}}{\sqrt{14}\sqrt{56}} = \frac{1}{14}$$

Condone used. $\overrightarrow{AB}$

Allow in terms of $|\overrightarrow{BC}|$ p , in which case previous B1 is implied

(m1)

$$-3 - 9p + 8 - 4p + 7 - p = 2$$

Reduce to linear or quadratic equation in p .

(m1)

$$p = \frac{5}{7}$$

(A1)

$$C \text{ is at } \left(9\frac{1}{7}, -2\frac{4}{7}, 9\frac{2}{7}\right)$$

(A1)

(6)

Alternative : using symmetry (i)

$$|\overline{AD}| = |\overline{BC}| = \sqrt{56}$$

$$\overline{AD} = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}$$

(B1ft)

$$|\overline{DC}| = |\overline{AB}| - |\overline{AD}| \cos \theta - |\overline{BC}| \cos \theta$$

Substitute values and evaluate

$$|\overline{AB}| - |\overline{AD}| \cos \theta - |\overline{BC}| \cos \theta$$

(M1)

$$|\overline{DC}| = \frac{10}{\sqrt{14}}$$

F on $\overline{AB}$ and $\cos \theta$

(A1ft)

$$|\overline{DC}| = p |\overline{AB}| \Rightarrow \frac{10}{\sqrt{14}} = p \sqrt{14}$$

Set up equation in p

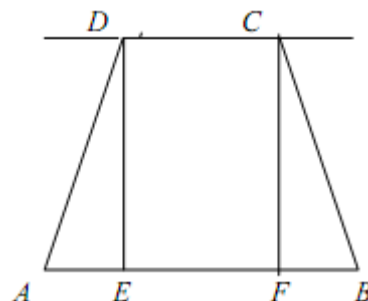
(m1)

$$p = \frac{5}{7}$$

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(A1)

$$C \text{ is at } \left(9\frac{1}{7}, -2\frac{4}{7}, 9\frac{2}{7} \right)$$



(A1)

(6)

Alternative using symmetry (ii)

$$|\overline{AD}| = \sqrt{56}$$

(B1ft)

$$|\overline{AE}| = |\overline{AD}| \cos \theta = 56 \times \frac{1}{14} = \frac{2}{\sqrt{14}}$$

Substitute values and evaluate

for $|\overline{AD}| \cos \theta$ *F on cos θ*

(M1)

(A1ft)

$$|\overline{AE}| = q |\overline{AB}| \Rightarrow \frac{2}{\sqrt{14}} = q \sqrt{14} \quad \text{and} \quad |\overline{AE}| = |\overline{FB}| \Rightarrow p = 1 - 2q$$

Set up equation to find p

(m1)

$$q = \frac{2}{14} \quad p = \frac{5}{7}$$

(A1)

$$C \text{ is at } \left(9\frac{1}{7}, -2\frac{4}{7}, 9\frac{2}{7} \right)$$

(A1)

(6)

[14]

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Q9.

$$(a) \quad \overline{AB} = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix} - \begin{bmatrix} 5 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$$

$$\pm (\overline{OA} - \overline{OB})$$

Co-ordinate form only is B0

Condone one component incorrect

B1

Line through A and B

$$\overline{OA} + \lambda \mathbf{d} \quad \text{or} \quad \overline{OB} + \lambda \mathbf{d} \quad \text{where} \quad \mathbf{d} = \overline{AB} \quad \text{or} \quad \overline{BA}$$

all in components and identified.

M1

$$\mathbf{r} = \begin{bmatrix} 5 \\ 1 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix} \text{ or } \mathbf{r} = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$$

$$\mathbf{r} \text{ or } \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ required}$$

Condone missing brackets on $\overrightarrow{OA}$ or $\overrightarrow{OB}$

A1

3

(b) (i) $5 - \lambda = -8 + 5\mu$

$$1 - 2\lambda = 5$$

$$-2 + 5\lambda = -6 - 2\mu$$

Clear attempt to set up and solve at least two simultaneous equations in μ and a different parameter. Allow in column vector form.

M1

$$\lambda = -2 \quad \mu = 3$$

One of λ or μ correct OE

A1

$$-2 + 5 \times -2 = -12 \quad -6 - 2 \times 3 = -12$$

Both equal -12 so intersect

Verify intersect, λ and μ correct or verify $(7, 5, -12)$ is on both lines; **statement required**

E1

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P is $(7, 5, -12)$

$$\text{CAO condone } P = \begin{bmatrix} 7 \\ 5 \\ -12 \end{bmatrix} \text{ OE}$$

and missing brackets

B1

4

(ii) $\overrightarrow{BC} = \begin{bmatrix} -8 + 5\mu \\ 5 \\ -6 - 2\mu \end{bmatrix} - \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} \text{ or}$$

$$\overrightarrow{CB} = \overrightarrow{OB} - \overrightarrow{OC}$$

B1

$$\begin{bmatrix} 3 \\ 6 \\ -15 \end{bmatrix} \cdot \overrightarrow{BC} = 0$$

Clear attempt at $\pm \overrightarrow{BP}$ or $\pm \overrightarrow{AB}$ or $\pm \overrightarrow{AP}$ in components sp with $\overrightarrow{BC} = 0$

M1

$$-36 + 15\mu + 36 + 135 + 30\mu = 0$$

Linear equation in μ using their $\overrightarrow{BC}$ and solved for μ . Condone one arithmetical or sign slip

m1

$$\mu = -3$$

A1

C is $(-23, 5, 0)$

CSO Condone column vector.

A1

5

[12]

Q10.

(a) (i)
$$\overrightarrow{AB} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$\pm (\overrightarrow{OA} - \overrightarrow{OB})$

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M1

A0 if answer as coordinates

A1

2

(ii)
$$\overrightarrow{OB} \cdot \overrightarrow{AB} = 3 \times 1 + 1 \times 0 + (-2) \times (-1) = 5$$

Evaluate to single value

M1A1

$$\cos \theta = \frac{\overrightarrow{OB} \cdot \overrightarrow{AB}}{|\overrightarrow{OB}| \times |\overrightarrow{AB}|}$$

Use formula for $\cos \theta$ with any 2 vectors and at least one of the corresponding

modulii 'correct'

M1

$$|OB| = \sqrt{14} \quad |AB| = \sqrt{2}$$

$$\cos \theta = \frac{5}{\sqrt{7 \times 2} \sqrt{2}} = \frac{5}{2\sqrt{7}}$$

CSO; AG so need to see intermediate step

$$\text{eg } \frac{5}{\sqrt{7 \times 2} \sqrt{2}} \text{ or } \frac{5}{\sqrt{28}}$$

A1

Alternative

cos rule attempted with cos B

(M1)

cos rule correct with cos B

(A1)

derive correct given form

(A2)

4

(b)

$$\mathbf{r} = \begin{bmatrix} 6 \\ 2 \\ -4 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$\overrightarrow{OC} + \lambda \overrightarrow{AB}$. Allow one slip

©2025 Exam Papers Practice. All rights reserved. M1

ft on $\overrightarrow{AB}$; needs $\mathbf{r}$ or $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

A1F

2

(c)

$$\overrightarrow{OD} \cdot \overrightarrow{AB} = \begin{bmatrix} 6 + \lambda \\ 2 \\ -4 - \lambda \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

M1

$$6 + \lambda + 4 + \lambda = 0$$

m1

$$\lambda = -5$$

ft on equation of line

A1F

D is $(1, 2, 1)$

CAO

A1

Alternative

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = a - c = 0$$

Let D be (a, b, c)

Scalar product evaluated and equated to 0

(M1)

$$a = 6 + \lambda, \quad b = 2, \quad c = -4 - \lambda$$

Use equation of line

(m1)(A1)

$$a + c = 2$$

$$a = 1 \quad b = 2 \quad c = 1$$

(A1)

4

[12]

Q11.

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(a) (i)

$$\overrightarrow{AB} = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ -3 \end{bmatrix}$$

M1 for $\pm (\overrightarrow{OA} - \overrightarrow{OB})$

M1A1

2

$$(ii) \quad (\mathbf{r} =) \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ -4 \\ -3 \end{bmatrix}$$

ft on $\overrightarrow{AB}$; OE

B1F

1

$$(b) \quad (i) \quad \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 5 \end{bmatrix}$$

μ found and verified or statement $\mu = -3$ satisfies all components

M1

$$1 + \mu = -2 \quad \mu = -3$$

$$-1 - 2\mu = 5 \quad \mu = -3$$

$\mu = -3$ alone B1

A1

ALTERNATIVE METHOD

$$\mu \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} \text{ which is satisfied by } \mu = -3$$

2

$$(ii) \quad \vec{PQ} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ -3 \end{pmatrix} - \begin{pmatrix} -2 \\ -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 + 2\lambda \\ 8 - 4\lambda \\ -4 - 3\lambda \end{pmatrix}$$

$\vec{PQ} = \vec{OQ} - \vec{OP}$ with $\vec{OQ}$ in parametric form in terms of λ (can be inferred later)

M1

$$\text{or } \begin{pmatrix} 6 + 2\lambda \\ 4 - 4\lambda \\ -7 - 3\lambda \end{pmatrix}$$

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A1

$$\begin{pmatrix} 4 + 2\lambda \\ 8 - 4\lambda \\ -4 - 3\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

$$\vec{PQ} \cdot \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \text{ with } \vec{PQ} \text{ in terms of } \lambda$$

(can be inferred later)

M1

$$(4 + 2\lambda) + (-2)(-4 - 3\lambda) = 0$$

linear expression in λ equated to 0

m1

$$\lambda = -1.5$$

ft on sign/arithmetic error in $\overline{PQ}$ or equation

A1F

Q is (-1, 11, 5.5)

CAO

A1

6

[11]

Q12.

(a) $AB^2 = (5 - 3)^2 + (3 - -2)^2 + (0 - 1)^2$

use $\pm (\overline{OB} - \overline{OA})$ in sum of squares of components
allow one slip in difference

M1

$$AB = \sqrt{30}$$

accept 5.5 or better

A1

2

(b) $\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} = 2 + 3 = 5$

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 $\pm \overline{AB} \cdot$ direction l evaluated
condone one component error

M1

5 or - 5

A1

$$\cos \theta = \frac{5}{\sqrt{30}\sqrt{10}}$$

F on either of candidates' vectors

B1F

use $|a| |b| \cos \theta = a \cdot b$; values needed

M1

$$\theta = 73^\circ$$

CAO
(condone 73.2, 73.22 or 73.22...)

A1

5

(c)
$$\overrightarrow{AC} = \begin{bmatrix} 5 + \lambda \\ 3 \\ -3\lambda \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 + \lambda \\ 5 \\ -1 - 3\lambda \end{bmatrix}$$

for $\overrightarrow{OC} - \overrightarrow{OA}$ or $\overrightarrow{OA} - \overrightarrow{OC}$ with $\overrightarrow{OC}$ in
terms of λ
condone one component error

M1A1

$$(2 + \lambda)^2 + 5^2 + (-1 - 3\lambda)^2 = 30$$

m1

$$10\lambda^2 + 10\lambda = 0$$

$$(\lambda = 0 \text{ or } \lambda = -1)$$

A1

$$(\lambda = 0 \Rightarrow (5, 3, 0) \text{ is } B)$$

$$\lambda = -1 \Rightarrow C \text{ is } (4, 3, 3)$$

condone $\begin{bmatrix} 4 \\ 3 \\ 3 \end{bmatrix}$

A1

5

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[12]

Q13.

(a) (i)
$$\overrightarrow{BA} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} - \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \\ 4 \end{bmatrix}$$

Attempt $\pm \overrightarrow{BA}$ (OA – OB or OB – OA)

M1A1

2

(ii)
$$\overrightarrow{BC} = \begin{bmatrix} 6 \\ 2 \\ -4 \end{bmatrix}$$

Allow $\vec{CB}$; or $\begin{bmatrix} -6 \\ -2 \\ 4 \end{bmatrix} = \vec{BC} \text{ or } \vec{CB} = \begin{bmatrix} 6 \\ 2 \\ -4 \end{bmatrix}$
May not see explicitly

B1

$$|\vec{BA}| = \sqrt{(-2)^2 + (-6)^2 + (4)^2} = \sqrt{56}$$

Calculate modulus of $\vec{BA}$ or $\vec{BC}$; for finding modulus of one of vectors they have used

B1F

$$\vec{BA} \cdot \vec{BC} = \begin{bmatrix} -2 \\ -6 \\ -4 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 2 \\ -4 \end{bmatrix} = -12 - 12 - 16$$

Attempt at $\vec{BA} \cdot \vec{BC}$ with numerical answer; or $\vec{AB} \cdot \vec{CB}$

M1

for -40 , or correct if done with multiples of vectors

A1

$$\cos ABC = \frac{-40}{\sqrt{56}\sqrt{56}} = -\frac{5}{7}$$

AG (convincingly obtained)

A1

Cosine rule: M1 attempt to find 3 sides

A1 lengths of sides

M1 cosine rule

A1F correct

$$\text{A1 rearrange to get } \cos ABC = -\frac{5}{7}$$

(ft on length of sides)

5

(b) (i) $\begin{bmatrix} 8 \\ -3 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 11 \\ 6 \\ -4 \end{bmatrix} \quad (\lambda = 3)$

$\lambda = 3$ verified in three equations

$$\text{M1 for } \begin{cases} 11 = 8 + \lambda \\ 6 = -3 + 3\lambda \\ -4 = 2 - 2\lambda \end{cases}$$

A1 for $\lambda = 3$ shown for all three equations



$$\lambda \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 11 \\ 6 \\ -4 \end{bmatrix} - \begin{bmatrix} 8 \\ -3 \\ 2 \end{bmatrix} \therefore \lambda = 3 \text{ M1A1}$$

SC: $\lambda = 3$ written and nothing else: SC1

M1A1

2

$$(ii) \begin{bmatrix} 2 \\ 6 \\ -4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

$\therefore$ same direction or same gradient or parallel

E1

1

$$(c) \vec{OD} = \vec{OC} + \vec{BA}$$

PI; $\vec{OD}$ = correct vector expression which may involve $\vec{AD}$

B1

$$= \begin{bmatrix} 11 \\ 6 \\ -4 \end{bmatrix} + \begin{bmatrix} -2 \\ -6 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \\ 0 \end{bmatrix} \quad D \text{ is } (9, 0, 0)$$

M1 for substituting into vector expression for $\vec{OD}$
NMS 3/3

M1A1

3

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[13]

Q14.

$$(a) \begin{bmatrix} 3 \\ -3 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} = 3 - 6 + 3 = 0$$

attempt at sp, 3 terms, added

M1

$= 0 \Rightarrow$ perpendicular

$= 0 \Rightarrow$ perpendicular seen (or $\cos \theta = 0 \Rightarrow \theta = 90^\circ$)

$$\text{Allow } \frac{3}{0} \text{ but not } \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} = 0$$

A1

2

(b) $8 + 3\lambda = -4 + \mu$

$$6 - 3\lambda = 2\mu$$

$$-9 - \lambda = 11 - 3\mu$$

$$\lambda = -2, \mu = 6$$

set up any two equations

M1

solve for λ and μ

m1 A1

verify third equation

substitute λ, μ in third equation

m1

intersect at $(2, 12, -7)$

CAO

A1

5

Alt (for last two marks)

substitute λ into l_1 and μ into l_2

(m1)

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intersect at $(2, 12, -7)$, condone

$(2, 12, -7)$ found from both lines

Note: working for (b) done in (a): award marks in (b)

(A1)

(c) $\vec{AP} = \begin{pmatrix} 6 \\ 12 \\ -18 \end{pmatrix}$

$$\vec{AP} = \pm \left\{ \text{their } \vec{OP} - \begin{pmatrix} -4 \\ 0 \\ 11 \end{pmatrix} \right\}$$

M1

$$AP^2 = 504$$

ft on p

A1F

$$AB^2 = 2AP^2$$

Calculate AB^2

M1

$$AB = 12\sqrt{7}$$

OE accept 31.7 or better

A1

4

[11]

Q15.

- (a) $\lambda = -4$ gives $P(-29, 42, -19)$ on l

Correct value of λ

B1

1

- (b) $\cos^{-1} \frac{1}{9}$ or 83.6° (or 84°) or 1.46 rads.

*ft from 3rd d.c. or by any other method
(e.g. scalar product)*

N.B. Mark lost if 6.4° is then offered as the answer

B1✓

1

- (c) (i) $\mathbf{n} = 3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

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B1

1

- (ii) Use of $\cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$

Must be direction vector of l and their $\mathbf{n}$

M1

$$\text{Nr.} = 45 \quad \text{Dr.} = \sqrt{50.9}$$

ft the "9" if necessary from (b)(i)

A1 A1

$$\theta = 45^\circ$$

CAO

A1

(d) Subst^d. $\begin{pmatrix} 3+8\lambda \\ 26-4\lambda \\ \lambda-15 \end{pmatrix}$ in $3x - 4y + 5z = 100$
 $3(3 + 8\lambda) - 4(26 - 4\lambda) + 5(\lambda - 15) = 100$

M1

Solving a linear eqn. in λ

dM1

$\lambda = 6$

CAO

A1

$\Rightarrow Q = (51, 2, -9)$

ft their λ in l

B1 ✓

4

(e) $PQ = \sqrt{80^2 + 40^2 + 10^2} = 90$
 ft

B1

Sh. Dist. = $90 \sin 45^\circ = 45\sqrt{2}$ or 63.6(4...)

M1

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A1 ✓

3

Or $p + m \perp n$ subst^d. into $\Pi \Rightarrow m = 9$

(M1)

$\Rightarrow R = (-2, 6, 26)$

$R =$ foot of perp^f. from P to Π

(A1)

$PR = \sqrt{27^2 + 36^2 + 45^2} = 45\sqrt{2}$
 ft

(B1 ✓)

(3)

Q16.

(a) (i) $\vec{AB} = \begin{bmatrix} 6 \\ 5 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix}$

M1

Penalise use of co-ordinates at first occurrence only

A1

2

(ii) $\begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow$ parallel

*Needs comment "same direction" Or "same gradient"
(Or by scalar product)*

E1

1

(iii) $\begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

M1

is satisfied by $\lambda = -4$

$\lambda = -4$ satisfies 2 equations

A1

2

(b) (i) l_2 has equation

$$r = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix} + \lambda \left[\begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix} \right] = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$$

Or

$$r = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix} + t \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$$

*M1 calculate and use
direction vector A1 all correct*

M1A1

2

$$(ii) \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix} = 4 - 4 = 0$$

Clear attempt to use directions of AC and l_2 in scalar product

M1A1

$\Rightarrow 90^\circ$ (or perpendicular)

Accept a correct ft value of $\cos \theta$

A1F

3

[10]

Q17.

$$(a) \quad (i) \quad \vec{OC} = 2 \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ -2 \end{bmatrix}$$

(Penalise coordinates once only)

B1

1

$$(ii) \quad \vec{AB} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$$

$\vec{OA} - \vec{OB}$ or $\vec{OB} - \vec{OA}$ or 2/3 correct components.

M1

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A0 for line AB

A1

$$(b) \quad (i) \quad AC^2 = (6-2)^2 + (4-4)^2 + (-1-2)^2 = 25$$

|Components of AC|

M1

$$AC = 5$$

AG

A1

2

$$(ii) \quad \vec{AB} \cdot \vec{AC} = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 0 \\ -3 \end{bmatrix} = 4 + 6 = 10$$

Clear attempt to use $\overline{AB}$ and $\overline{AC}$

M1

ft $\overline{AB}$ from a(ii) and/or $\overline{AC}$ from b(i)

A1F

$$3 \times 5 \times \cos \theta = 10$$

Use of $|a| |b| \cos \theta = \mathbf{a.b}$

With one correct $| |$ and $\mathbf{a.b}$ evaluated

M1

$$\theta = 48.189 \approx 48^\circ$$

CAO (AWRT)

A1

4

Alternative: use of cos rule
Find 3rd side + use cos rule

(M2)

ft on previously found vectors

(A1F)

CAO (AWRT)

(A1)

(c) $\overline{BP} = \begin{bmatrix} \alpha - 3 \\ \beta - 2 \\ \gamma - 1 \end{bmatrix}$

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B1

$$\begin{bmatrix} 4 \\ 0 \\ -3 \end{bmatrix} \cdot \overrightarrow{BP} = 0$$

Their $\overline{BP}$

M1

$$4\alpha - 3\gamma - 15 = 0$$

AG convincingly obtained

A1

3

[12]

Q18.

(a) $|\vec{AB}| = \sqrt{(2-1)^2 + (-1-4)^2 + (3-2)^2}$
OE

M1

$= \sqrt{27} = 3\sqrt{3}$
AG; convincingly obtained

A1

2

(b) $\begin{bmatrix} 1 \\ -5 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 1 + 5 + 1$

Their $\vec{AB} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

M1

Their scalar product $= \begin{vmatrix} 1 \\ -1 \\ 1 \end{vmatrix} \cdot 3\sqrt{3} \cos \theta$

M1

$7 = 3 \cdot \sqrt{3} \cdot \sqrt{3} \cos \theta \quad \cos \theta = \frac{7}{9}$
AG; convincingly obtained

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A1

3

(c) (i) $\vec{OP} = \begin{bmatrix} 2+p \\ -1-p \\ 3+p \end{bmatrix} \quad \vec{AP} = \begin{bmatrix} 1+p \\ -5-p \\ 1+p \end{bmatrix}$
Finding $\vec{AP}$

M1A1

$\vec{AP} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = (1+p) - (-5-p) + (1+p)$

m1

$= 7 + 3p$

AG

SC working with λ instead of p giving

$$7 + 3\lambda = \frac{3}{4}$$

A1

4

(ii) $7 + 3p = 0$

M1

$$p = -\frac{7}{3} \quad P \text{ is } \begin{pmatrix} -\frac{1}{3} & \frac{4}{3} & \frac{2}{3} \end{pmatrix}$$

Allow column vectors and decimals

A1A1

3

[12]



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Examiner reports

Q2.

There was a wide range of responses to this question and relatively few students gained full marks, although about 25% of the entry did score 11 or more of the 15 marks available.

Part (a)(i). Most students answered this correctly, others making the occasional sign error.

Part (a)(ii). Most students found either the vector $\overrightarrow{BC}$ or $\overrightarrow{CB}$ correctly, but some students mixed the two up and others made sign errors. Most students did work with the vectors $\pm \overrightarrow{AC}$, $\pm \overrightarrow{BC}$ so focussing on the required angle, although some found the obtuse rather than the acute angle; they could gain full marks if they explained why their result was negative. Few students tried to work with the wrong vectors, $\overrightarrow{OA}$ and $\overrightarrow{OB}$ being the usual choice of those who did. The scalar product based formula for $\cos \theta$ is well known, but some students did approach the problem using the cosine rule with most being successful by this method. Some students only scored 3 of the marks as they gave insufficient justification as to how their expression became that given in the question.

Part (b). Most students knew what was required here although many scored only 1 mark due to poor notation. $\overrightarrow{AC} = \mathbf{a} + \lambda \mathbf{d}$ or line $AC = \mathbf{a} + \lambda \mathbf{d}$ or similar were common rather than $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$.

Part (c)(i). Many students showed they understood the question and found a vector equation for line BD and solved simultaneously with that of line AC to find values for the parameters. The fairly common error was to use the vector $\overrightarrow{BD}$ itself as the equation of the line, which resulted in no credit. Some students omitted their parameter in the third equation and so found $p = -6$ without attempting to find the value of their parameter. Many did find the value of p correctly, with relatively few making mistakes in the algebra although some had difficulty in solving $\frac{1}{2}p = -6$. Those few who chose point D as a point on line BD made the algebra unnecessarily complex and often solved incorrectly.

Part (b)(ii). The expected approach to showing $ABCD$ is a rhombus was to find the lengths of the four sides. Although some students started to do this either by finding the vectors first, or going directly to their moduli, few attempted all four and so could gain no credit. If they had made an earlier mistake 2 marks were still available, but many students seem to give up if they weren't getting equal lengths. Those who just stated all lengths are

7 (from part (a) and $|\overrightarrow{BC}|$) or showed no detail of where 7 came from for each side, gained no credit. Those relatively few students who found the scalar product $\overrightarrow{AC} \cdot \overrightarrow{BD} = 0$ and claimed the diagonals are perpendicular hence the shape is a rhombus, scored 2 of the 4 marks as they didn't consider that the diagonals also needed to bisect each other. Few students drew a diagram for this part of the question, which might have aided their solution.

Q3.

Although most students scored well on parts (a) and (b), many made little further progress and generally the question was not done well. However, about 15% of the entry did score 10 or more of the marks available. There were few fully correct solutions to part (d). Notation was often poor and many students did not make it clear which vectors they were working with.

- (a) Most students answered this correctly. Many found $\lambda = -1$ explicitly or wrote a correct vector expression in which it was implied. Most demonstrated that this value satisfied the three components although a few students failed to carry out this step. The other rarely seen error was to find $\lambda = +1$. It was good to see some students writing a clear conclusion to their solution, although this was not required.
- (b) Most students knew they had to find the vector $\overrightarrow{AB}$ and apart from the occasional sign error they did so successfully. However, some thought this was the equation of the line. Many others had the correct form of the line but omitted $\mathbf{r} =$, or wrote $line =$ or something similarly inappropriate for the vector equation of a line.
- (c) Although most students knew they were to use the equation of the line and equate a scalar product with vector $\overrightarrow{AB}$ or $\overrightarrow{AD}$ to zero, many used $\overrightarrow{OD}$ rather than $\overrightarrow{CD}$ and so could gain no further marks. Those who tried to find vector $\overrightarrow{OD}$ often made a sign or coefficient error, so could only score the method marks in finding the value of μ and the coordinates of D . Some got as far as $17 + 17\mu = 0$ and concluded $\mu = 1$.
- (d) Many students did not attempt this part of the question. Success here would have been more likely if a correct diagram had been used, but most of those who attempted this part of the question did not draw one and many had the right angle in the wrong place, at C instead of at D . Many then used the sides AC and AD of the triangle ACD to attempt to find the area whereas the base and height are AD and CD respectively. Some students who chose this route, especially if they had an error in the coordinates of D , continued working with very unfriendly fractions and attempts at finding moduli before abandoning the attempt for no credit. The key to a successful solution lies in realising that CD is the height of both triangles ACD and ACE and as such there is no need to actually find its length, although students found that CD cancels if they did this correctly. Some students did give a concise and correct solution for both the possible positions of point E .

Q4.

Well over half of the candidates scored the first 6 marks, showing knowledge and competence in finding a vector from coordinates and using the appropriate vectors to find an angle between a vector and a line.

In part (a)(i), most candidates showed they knew how to find the vector $\overrightarrow{AB}$, with relatively few doing the subtraction the wrong way round. Some candidates made arithmetic slips.

In part (a)(ii), virtually all candidates showed that they knew how to use a scalar product to find an angle between two vectors, but some candidates chose the position vectors of points A and B rather than vector $\overrightarrow{AB}$ and the direction vector of the line; they could score one mark if they used the formula correctly. A few equated to the sine rather than the cosine of the angle. Some candidates dropped a sign when copying the vector $\overrightarrow{AB}$, so the scalar product then often became zero; these candidates correctly interpreted this result as 90° , and gained a 1 mark credit for this. Many candidates, however, went on to get the right answer. Many unnecessarily manipulated their expression for the cosine using surds before finally doing the appropriate calculation on their calculator.

The key to success in a question such as this part (b) is to draw a diagram to aid geometrical thinking. Relatively few candidates did so, and of those that did some had the given right angle in the wrong place. To progress further, candidates needed to find the

vector $\overrightarrow{BC}$ in parametric form and use it in a scalar product with vector $\overrightarrow{AB}$ in order to find the value of the parameter that leads to the result. Those who did not do this could score no further marks.

Of those candidates who showed they knew in principle what to do, many made an error in their arithmetic so could not get to a correct result for the parameter. It so happened that many candidates used vector $\overrightarrow{OC}$ rather than $\overrightarrow{BC}$, made an error and got what appeared to be the right value for the parameter and then completed to get the right answer; however they were given no credit. Some candidates, who got as far as successfully finding the coordinates of point C , could not see how to use the geometry of the rectangle to find the coordinates of D . Some went back into parameters and scalar products, whilst others tried to equate the lengths of the sides, all to no avail. Relatively few achieved full marks on this question.

Q5.

This was another high-scoring question for many candidates, especially since the three parts were essentially completely independent of each other's results. Part (a) was straightforward and routine, though rather a lot of candidates lost marks by failing to give the acute angle required, subtracting the correct one from 90° , mixing up sines and cosines in the working, missing the minus sign in front of the "4" in the second of the two given normal vectors, or making numerical slips in adding up the three squares in each of the moduli.

Q6.

In part (a), most students set up the simultaneous equations required correctly, with many of these spotting the solution $\mu = -1$ immediately and thus finding the value of λ easily. Other students used conventional simultaneous equations methods, which took rather longer but could still be done successfully. Most then went on to use the third equation to show that $q = 4$, with relatively few assuming $q = 4$ and showing that it works. Most found the coordinates of the point P correctly, with the occasional coefficient error, although most gave their solution as a column vector rather than coordinates, which was condoned this time.

In part (b), most students knew that they were to show a scalar product came to zero, and most correctly choose the direction vectors and demonstrated this was the case. There was no credit for just stating the result, and students who failed to state the conclusion to their scalar product = 0 were also penalised.

Many students misunderstood the notation in part (c)(i), not realising they were being asked to find the square of the length AP . Some students squared their coordinates of point P , before subtracting those of point A , and leaving it as their answer, whereas some multiplied the corresponding coordinates. Some students showed some understanding in getting an answer of (16, 0, 4) but left the answer as a column vector and not a length squared, so got no credit. Some students who had misunderstood part (c)(i) found the length of AP correctly, when attempting part (c)(ii).

In order to make any progress with part (c)(ii), students needed to express the vector $\overrightarrow{BP}$ or $\overrightarrow{AB}$ in terms of a parameter, find the length of the vector and use the properties of the isosceles triangle to set up an equation. There were some good attempts at this stage although rather fewer went on to complete fully successfully. Errors were made in expanding expressions such as $(a + bp)^2$ or in the collection of terms, although many did get to, and attempted to solve, a quadratic equation as required and gained credit for

using a correct method. Some students attempted to use the scalar product $\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$ but this just reduces to $0 = 0$ if done correctly. There is an alternative method avoiding the use of a parameter, based on equating vector $\overrightarrow{BP}$ to a multiple of the direction vector of line l_2 , but this was rarely seen.

Q7.

As already mentioned in the introduction, this question proved to be very demanding and frequently candidates offered no attempt. Even though the three marks available in parts (a) and (b) were relatively straightforward, hardly any students managed to acquire all of them.

This was because in part (b) almost all students undertook the expected method, gaining a zero scalar product, but then failed to consider that some explanation would be required as to why a successful perpendicularity test established parallelism.

Students arguably could be criticised for spurning some relatively straightforward marks earlier on in this paper, but not in the last part of this final question: it was always intended to discriminate and to stretch the most able. Most students did not know where to begin and so made no attempt. Most of the rest presented almost unintelligible amounts of any vector method they could think of at the problem, almost invariably without any explanation or structure to their working. As a result, there were three or four marks which we were prepared to give for useful introductory generalised points, vectors or distances, but hardly anyone could manage even these. It did not help when many, of the few who did try to do something here, thought that equal distances PQ and PR meant equal vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

Q8.

The first part of this question was found to be fairly straight forward by most candidates, but then the second part defeated most; although some fully, or almost fully correct solutions were seen.

Part (a)(i) was answered well, although some candidates made arithmetic errors. A few lost a mark for giving their answer in coordinate form and a few found $\overrightarrow{OA} - \overrightarrow{OB}$.

In (a)(ii) most candidates worked with the correct vectors, although some used the position vectors of A and/or B . The scalar product was usually found correctly and the formula for finding the cos in e of the angle was well known. However, having set up the correct equation, some candidates equated the product of $\sqrt{14}\sqrt{14}$ to $2\sqrt{14}$, so found an incorrect angle. Some made an error in finding a modulus.

In part (b)(i) almost all found the vector $\overrightarrow{OD}$ correctly. However very few candidates gained the mark for the equation of the line as most wrote $l_2 = ..$ instead of $\underline{r} = ..$ so they didn't actually have a vector equation of the line.

In part (b)(ii) the problem required the candidates to both devise a strategy for its solution and carry it out.

The key to the solution was realising that the moduli of $\overrightarrow{AD}$ and $\overrightarrow{BC}$ were equal, which could then be used directly or in setting up an equation involving equal angles. However, many candidates either took the vectors $\overrightarrow{AD}$ and $\overrightarrow{BC}$ to be equal, or that $\overrightarrow{AD} = -\overrightarrow{BC}$. They got a wrong result for the coordinates of C very quickly from this assumption. Those few candidates who got as far as finding $p = 5/7$ or 1 usually opted for the latter root (some stating μ had to be an integer) and apparently didn't think about the geometry of

the trapezium.

Those who did have an appropriate strategy often made an error in setting up an equation and/or made an algebraic or numerical error in solving it.

Q9.

In part (a) most candidates had the line correct in terms of a point on the line and its direction vector but the notation was often poor, with many not giving an equation by omitting $\mathbf{r} =$ or stating line = or similar. Some made sign errors in finding their direction vector.

In (b)(i) most candidates knew they needed to equate the two vector equations of the lines and solve simultaneously for λ and μ . Some candidates failed to change parameter and had three equations in μ and could make no further progress. Most candidates noticed that the equations had been set up so that it was easy to find λ first, and although they could calculate the intersection point just from the value of λ , most did find μ before doing so.

Most candidates attempted to check that all the equations were satisfied but many failed to bring this to a conclusion, just saying they were all satisfied, or even just $12 = 12$ from the third equation, without stating the implication; it was necessary to say 'therefore the lines intersect', or similar, to gain the mark. Most did find the intersection point correctly, although several made sign errors. The few who had an error in their solution to the simultaneous equations, appeared not to question their solution when finding the third equation didn't check.

Attempts at part (b)(ii) of the question ranged from no attempt, or just a superficial diagram with no progress, to about 15% of the candidates giving a fully correct solution. Most knew they were to equate a scalar product to zero, but many chose at least one wrong vector. Those who drew a diagram usually chose the right vectors, but often used $\overrightarrow{OC}$ instead of $\overrightarrow{BC}$ in their calculation. Many who did attempt to calculate $\overrightarrow{BC}$ correctly made a sign or coefficient error which meant they could get neither the correct value of μ nor the correct coordinates for point C. No credit was given for expanding the scalar product until it was in terms of one parameter and thus soluble. A few candidates equated each term of their scalar product to zero, in order to find values for x , y and z or whatever they had called the coordinates of point C. The other fairly common misunderstanding

was just to use a parameter p , as in $\overrightarrow{BC} = (p-4 \ p+1 \ p-3)^T$ rather than the equation of the line, to parameterise the point C. A few candidates attempted to use Pythagoras Theorem to solve this problem, with little success, although some did give a fully correct solution.

Q10.

Most candidates used an appropriate method in part (a)(i) to find the requested vector, with some making a sign error, particularly with the z component, where $-1 - -2 = -3$ was common. Some candidates expressed the vector as if it were coordinates and were penalised. For part (a)(ii), although many candidates did choose the correct vectors for the scalar product and had them clearly identified, some worked with the position vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$. Of these, some proceeded to get their own incorrect answer, whilst others were determined to get to the given result for $\cos \theta$, without apparently considering that they had made an error.

Some candidates had the negative of one of the vectors they should have been working with and just dropped the resulting minus sign. Some candidates, who were otherwise correct, were denied a mark for not showing how $\sqrt{14}\sqrt{2}$ became $2\sqrt{7}$. For part (b), most candidates knew the $\mathbf{a} + \lambda\mathbf{b}$ form for the vector equation of a line, although few used the correct notation of $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ and just called it $\mathbf{l} = \mathbf{a} + \lambda\mathbf{b}$, which was penalised. Some candidates had the position vector $\mathbf{a}$ and the direction vector $\mathbf{b}$ reversed.

There were relatively few successful attempts at part (c). Many candidates knew they had to equate a scalar product to zero, but attempted $\vec{OD} \cdot \vec{OC}$ or used the parallel vectors $\vec{AB}$ and $\vec{CD}$. A diagram would probably have helped to identify the correct vectors to use. Some did use $\vec{CD}$ in place of $\vec{AB}$ which was correct, but candidates were more prone to error in solving the resulting quadratic equation. However, some well presented fully correct solutions were seen.

Q11.

This question also separated clearly into two parts, with most candidates scoring well in part (a) and part (b)(i) but few making substantial progress in part (b)(ii).

Part (a)(i) was usually correct but with the occasional sign error or the position vectors the wrong way round. In part (a)(ii), some candidates chose to give one of the position vectors as their direction vector.

There were several approaches to part (b)(i), most of which were acceptable, the important thing here being to give a convincing argument that the given point does lie on the line. This was best done by finding the value of μ from one component equation and showing that it satisfies the others.

Candidates whose working was unclear, with $\mu = -3$ appearing somewhere within it, usually only scored 1 mark. Candidates should always state which vectors they are

working with in terms of letters, such as $\vec{AB}$, rather than giving numerical values which might not be recognisable. This was particularly the case in part (b)(ii), where it often was not clear what candidates were trying to do as they had not identified the vectors with which they were working. However, it was apparent that, of those who did make a

recognisable start, some tried to work with vector $\vec{OQ}$ instead of $\vec{PQ}$ and then often went on to take the scalar product with the wrong vector. Some candidates tried to work with a

vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, again often without saying what it represented, and, although a few eventually expressed x , y and z in terms of λ , by then many had made a mistake and it became difficult to follow how they had got to their incorrect result. The use of the parametric equation of the line to solve a problem of this nature appeared not to be well known.

Q12.

In part (a) relatively few candidates went directly to the distance formula, with most finding the vector $\vec{AB}$ first. Some stopped there and thus gained no credit. Many candidates obtained the correct result, with any errors made usually occurring in misread signs or coefficients.

For part (b) many candidates identified the correct vectors to use and found the required angle correctly. The common error was to use the vectors $\vec{OA}$ and $\vec{OB}$ and OB ; such candidates gained partial credit for knowledge and use of the appropriate formula. Other candidates had a mix of vectors, and for some of these it was not possible to see what was intended and where they had got their vectors from. Some candidates used one pair of vectors for a scalar product and another pair for the moduli in their formula for $\cos \theta$.

There were few complete good answers to part (c), although some candidates were very inventive in their solutions, showing good geometrical insight. Many candidates showed little understanding of the geometry, with those who drew a diagram usually being the more successful. Many candidates equated the vectors $\vec{AC}$ and $\vec{AB}$ and went on to find B and C were the same point, without apparently realising any error had been made. Some candidates attempted to find the intersection of line l with line AB , again not noting the intersection point was B . Some candidates did take the expected approach and found $\vec{AC}$ in terms of parameter λ , but then didn't know what to do with it. Some who did equate the distances, made an error in their algebra, and ultimately few fully correct solutions were seen.

Q13.

In part (a) most candidates did attempt to find $\vec{BA}$ although some did the calculation for $\vec{AB}$, and despite being asked for $\vec{BA}$ in part (a)(i) many candidates proceeded to do the scalar product in (a)(ii) using $\vec{AB} \cdot \vec{BC}$. Such candidates often slipped a minus sign into their final answer rather than look for an error, either unexplained or with a comment like "it's the other angle". Some candidates used the position vectors, some finding a scalar product and multiplying the magnitudes of all three of $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$. A few candidates used the cosine rule, which is a valid method, but the attempts tended to be error prone.

Part (b) was generally done well, although the amount of explanation offered varied considerably. Those who solved one component equation for $\lambda = 3$ needed to verify it in both the other equations. Those who just saw that $\lambda = 3$ needed to demonstrate that this

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did give vector $\vec{OC}$ (many did), or the equivalent form of $\begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 9 \\ -6 \end{bmatrix}$. In part (b)(ii) a comment was expected to accompany $\lambda = 2$ or $\lambda = -2$, the latter being accepted, about the direction vectors being the same or, as a minimum, just stating this shows they are parallel. Simply stating the direction vectors are the same without showing it was not accepted.

There were several approaches that could be made to the problem in part (c). The correct answer can just be written down and some candidates did this, it coming from $\vec{OA} + \vec{BC}$. Very few candidates drew a diagram or made use of previous parts of the question, which is perhaps why many sign errors were made in the directions of the vectors, and correct directions are essential to this problem.

The coordinates of D as $(13, 12, -8)$ was a common wrong answer. Although it lies on line l , it does not give AD as parallel to BC . Some candidates attempted to use intersecting lines, which does lead to the correct answer, although many candidates made an error in their algebra. Another possibility used by some candidates was to equate the lengths of the sides of the parallelogram. Some candidates, who had achieved highly on

the rest of the paper, pursued their ideas here relentlessly, rather than stopping and rethinking the problem after a couple of sides of working in which they had made an error.

Q14.

In part (a), most candidates were successful, both showing that they knew the condition for two lines to be perpendicular and demonstrating that the scalar product was zero. A few failed to state the conclusion after finding a zero scalar product. Quite a number of candidates started their solution to part (a) as if they were answering part (b), with some stating the conclusion that the lines intersect and so are perpendicular. Although this gained no credit for part (a), it was credited in part (b).

In part (b), most candidates knew they were to set up simultaneous equation and solve them which most did successfully. Both elimination and substitution techniques were seen, the latter usually requiring rather more work. The common errors were in the signs. Some candidates failed to verify that an independent third equation was also satisfied by their solutions or that their intersection point indeed lay on both lines. Some, who attempted verification and found that it did not work, did not then check their own working for an error; they simply concluded that the lines did not intersect, with some adamantly stating that the question was wrong.

Fully correct solutions to part (c) of the question were rare. Many candidates got no further than finding the vector $\vec{AP}$ from their previous result, then taking the question to mean $\vec{AP} = \vec{BP}$ rather than that the lengths of these two vectors were the same. Those candidates who did realise that the question was about lengths of vectors and drew a diagram were generally successful. Others were determined that they could find the point B by using an arbitrary point on the line l , often assuming $\lambda = 0$. Many concluded that A and B were the same point and so the distance between them was zero.

Q15.

Parts (a) to (d) of this question were usually very well done, and candidates were able to score a lot of the marks. The big surprise came in part (e) when so few arrived at the correct final answer. In almost all cases, a simple diagram would have helped enormously, and they would then have seen that a simple bit of basic trigonometry would have done the trick. In most cases, candidates seemed to be working with randomly-selected vectors from the first few parts of the question and trying to insert them into some (often half-) remembered vector formula.

Q16.

In part (a)(i) the vector $\vec{AB}$ was often just written down with no working shown, although sign errors were permitted for the method mark.

$$\begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

In part (a)(ii) many candidates just wrote $\begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ with no explanation as to how this answered the question. A reference to having the same direction vector, or gradient, was expected.

There was a mix of responses to part (a) (iii). The most convincing were those in which candidates set up two equations and derived the value of λ from one of them and showed it satisfied the other. Some made a comment on the third equation, $-1 = -1$ although this

wasn't required. Some candidates wrote $\begin{bmatrix} -4 \\ -4 \\ 0 \end{bmatrix} = -4 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ without any convincing explanation of why it showed point D lies in the line, although some partial credit was often awarded, depending on the explanation.

For part (b) (i) most candidates showed they knew they had to find the direction vector $\overrightarrow{MD}$, although some made numerical errors in doing so. Many candidates omitted to write 'r =' in writing down the equation of the line, with $l_2 =$, or *line is* or just the vectors, being common poor notations which were penalised. For part (b) (ii) virtually all candidates knew a scalar product was involved, although some chose incorrect vectors, using points rather than directions. Some who used the correct vectors and correctly evaluated the scalar product as 0, still calculated the moduli of the vectors, although all knew a zero value implied 90° . A correct angle often followed an arithmetic slip in the scalar products and this was given partial credit.

Q17.

Most candidates gained some marks on the vectors question, particularly in parts (a) and (b)(i). Some candidates persist in using coordinate notation instead of components, and this was penalised.

In part (a) although the vector $\overrightarrow{OC}$ could just be written down, some candidates subtracted the zero vector before doubling the result. Virtually all candidates knew how to find the vector $\overrightarrow{AB}$, although sign and arithmetic errors were fairly common.

Part (b)(i) Most candidates did this successfully by first finding the vector $\overrightarrow{AC}$ and then calculating its modulus. Those candidates who did not show why they were squaring 3 and 4 to get the result, were penalised. In part (b)(ii), most candidates showed that they can calculate an angle between two vectors, but rather fewer showed they could choose the correct two vectors with the correct directions. The angle was given in the question as BAC and so candidates were expected to work with $\overrightarrow{AB}$ and $\overrightarrow{AC}$ or $\overrightarrow{BA}$ and $\overrightarrow{CA}$. Many candidates had a direction incorrect on one of these vectors, whereas many more used position vectors, or vectors where it just was not clear where they had got them from. Some candidates calculated the lengths of the sides of the triangle and used the cosine rule, often successfully.

Part (c) Answers ranged here from a well set out high quality demonstration of the result, to considerable illicit manipulation of vectors to get the given result. Most candidates knew a zero scalar product was involved, but many were not at all clear in what way. Here, and also in part (b), some candidates gave their evaluated scalar product in the form of a column vector, rather than as a scalar which is incorrect and was penalised. Some just worked with the vector $\overrightarrow{OP}$ but still managed to "derive" the result.

Q18.

Part (a) This was usually done well with most candidates obtaining the components 1, 5, 1 and showing the connection to $\sqrt{27}$.

Part (b) Here there was an unexpected lack of knowledge of the scalar product. Although $|a| |b| \cos \theta = a \cdot b$ was often quoted, many candidates left their scalar product in a column and few could convincingly show where $7/9$ came from.

Part (c)(i) This was seen more so here, with scalar products written as columns and the given answer just written down. Candidates were expected to find $\vec{AP}$ using $\vec{OP} - \vec{OA}$ and relatively few did it convincingly.

Part (c)(ii) Most candidates understood that they needed to put $7 + 3p = 0$ and went on to solve for p and to find the coordinates correctly, although some made arithmetical errors.



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